Year 12 Maths - Working with data

Similar questions - Averages, spread and outliers

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The number of text messages sent in one day by each of 40 students is summarised below.

Messages	0	1–5	6–10	11–20	21–50
Frequency	3	9	14	10	4

(a) Write down the class that contains the median and the class that contains the upper quartile. **[2]**

(b) Estimate the mean and the standard deviation of the number of messages. **[4]**

(c) A new student who sent 140 messages is added to the data. Without calculating, state the effect on the median and on the mean, and say which average is better to use. **[3]**

2 The heights, in cm, of 15 seedlings are listed below.

12 14 17 20 23 23 25 26 28 29 31 32 34 40 78

(a) Find the median and the interquartile range. **[3]**

(b) A value is an outlier if it is more than $1.5\times$IQR above Q_3 or below Q_1. Show that 78 is an outlier and that there are no others. **[3]**

(c) Calculate the mean of all 15 heights and the mean with the outlier removed. Explain why they differ so much. **[3]**

(d) Describe how the outlier would be shown on a box-and-whisker plot. **[1]**

Question 1

(a) $n = 40$: median is between the 20th and 21st values, which lie in the class 6–10 ($3+9=12$, $12+14=26$). Upper quartile is the 30th value: cumulative frequency after 11–20 is 36, so the class 11–20

(b) Midpoints $0, 3, 8, 15.5, 35.5$: $\bar{x} = \dfrac{0+27+112+155+142}{40} = \dfrac{436}{40} = 10.9$. $\sum fx^2 = 0+81+896+2402.5+5041 = 8420.5$, so $\sigma = \sqrt{\frac{8420.5}{40} - 10.9^2} = 9.58$ (using the midpoints)

(c) The median moves up by only a little (to the 21st value of 41, still in 6–10); the mean increases a lot (by roughly 3) because 140 is far from the rest. The median is better because it is less affected by extreme values

Question 2

(a) Median is the 8th value, 26. Lower half $12, 14, 17, 20, 23, 23, 25$ gives $Q_1 = 20$; upper half $28, 29, 31, 32, 34, 40, 78$ gives $Q_3 = 32$. IQR $= 12$

(b) $Q_3 + 1.5 \times 12 = 50$, and $78 > 50$, so 78 is an outlier. $Q_1 - 18 = 2$ and no value is below 2. $40 < 50$ is not an outlier

(c) Mean of all 15: $\dfrac{432}{15} = 28.8$; without 78: $\dfrac{354}{14} = 25.3$ (to 3 s.f.). The outlier is far above the others and pulls the mean up

(d) As a cross (or point) beyond the end of the upper whisker, which stops at the largest value within the fence, 40